CLASS 10 · JEE ADVANCED · INSTITUTE TOP 10 VS OUR TOP 10

# Pragya G10 (EXI) Achievers

Part Test-1 · Paper 2 · where our toppers fall behind the institute's toppers

**Exam date** 09 Oct 2026
**Campus** HYD/Nallakunta DD Colony
**Compared** institute top 10 vs our top 10

On average the institute's top 10 scored ≈ **132.0** and our top 10 scored ≈ **129.7** – a gap of ≈ **2.3 marks** per student: **1.8** in Maths and **0.9** in Physics.

Maths 67% of the gap | Physics 33% of the gap

**7 of the institute's top 10 are our students** (program rank 1, 2, 4, 5, 6, 7, 8), so this compares the institute's other 3 toppers with our next 3. Scores are rebuilt from each question's right / wrong / skipped counts with the paper's marking; approximate (≈) because multiple-correct questions give partial marks – our official average is 128.9.

**≈ 132** / 176
institute top 10 – average score

**≈ 130** / 176
our top 10 – average score

**5** questions
where 2+ more of the institute top 10 got it right

**4** questions
where 2+ more of ours got it right

## 1 Subject by subject

Average marks per student, and how the answers split into right, wrong and skipped. ■ Institute top 10 ■ Our top 10

| SUBJECT | AVERAGE SCORE | | GAP | RIGHT | WRONG | SKIPPED |
|---|---|---|---|---|---|---|
| **Maths**<br>out of 60 | ≈ 43.7<br>≈ 41.9 | | **1.8**<br>per student | **77%** vs **75%** | **17%** vs **18%** | **6%** vs **6%** |
| **Physics**<br>out of 56<br>= 60 minus Q25 (4) | ≈ 38.3<br>≈ 37.4 | | **0.9**<br>per student | **67%** vs **66%** | **22%** vs **26%** | **11%** vs **8%** |
| **Chemistry**<br>out of 60 | ≈ 50.0<br>≈ 50.4 | | **−0.4**<br>per student | **85%** vs **85%** | **14%** vs **13%** | **2%** vs **2%** |

"Out of" = the subject's marks on this paper minus the questions left out by the final key (marks of each left-out question in brackets). Percentages are of all answers in that subject (questions × students), institute first, ours second.

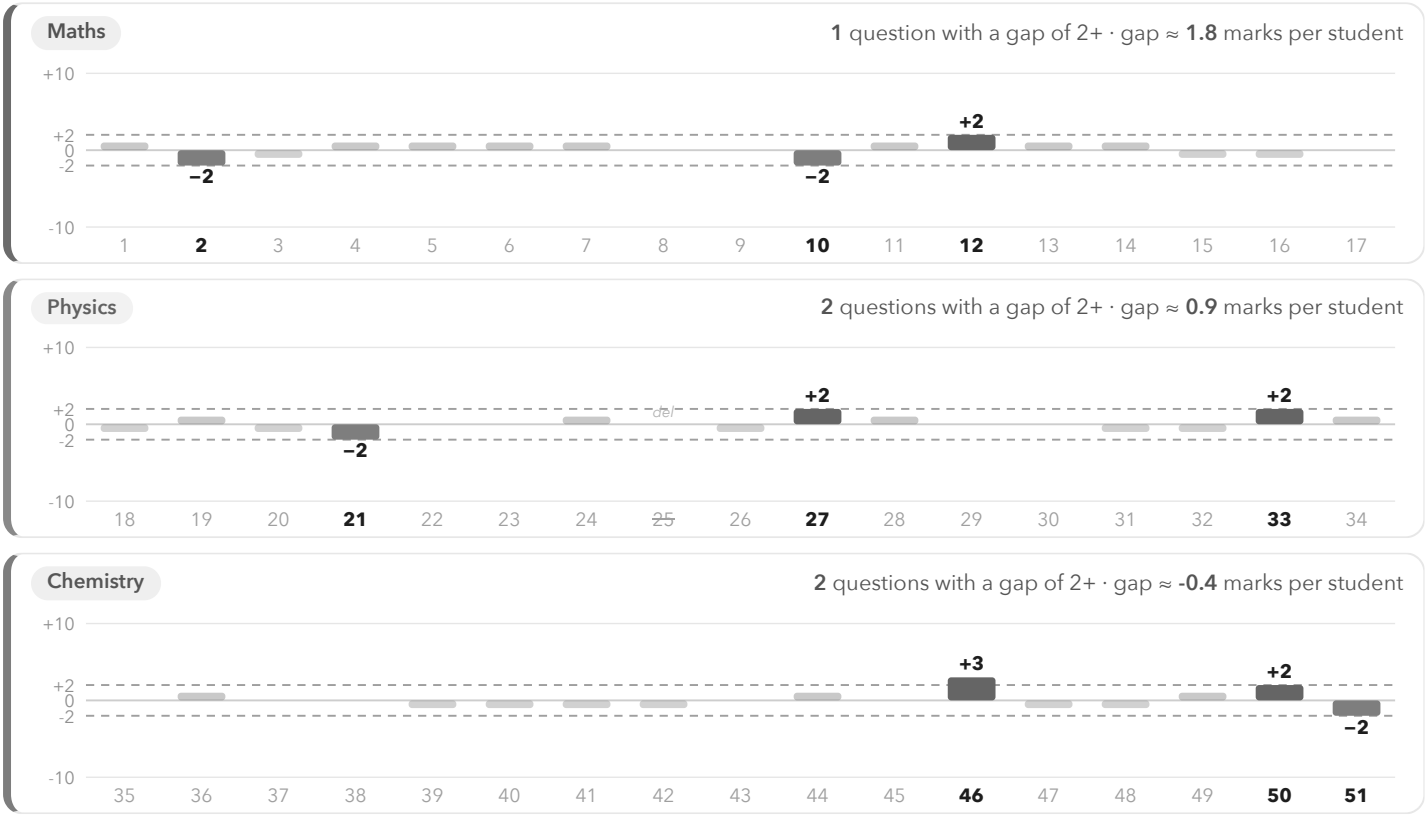
### Key takeaways

1. **7 of the institute's top 10** are our own students (program rank 1, 2, 4, 5, 6, 7, 8), so the two groups mostly overlap: at most 3 answers can differ on any question, so the gap line is set at 2 instead of 4.
2. **Maths** is where we fall furthest behind: ≈ **1.8** marks per student (67% of the total gap).
3. **5 questions** were answered correctly by at least 2 more of the institute's top 10 than ours. Biggest: Q46 (5 vs 2), Q12 (8 vs 6), Q27 (8 vs 6).
4. Where we did better: Q2 (5 vs 3), Q10 (10 vs 8), Q21 (6 vs 4), Q51 (5 vs 3).

## 2 Question-by-question gap

For each question: how many more of the institute's top 10 got it right than ours. Bars above the line = the institute did better; below = we did better. Gaps of 2+ (dashed line) are labelled.

■ Institute top 10 got more right ■ Our top 10 got more right ---- Gap of 2




## 3 Chapter gaps

Average marks per student on each chapter's questions, institute top 10 vs ours. Subjects are ordered from the biggest gap to the smallest; within each subject, largest chapter gap first.

| CHAPTER | QS | INSTITUTE VS OURS | GAP PER STUDENT | GAP QUESTIONS |
|---|---|---|---|---|
| **1. Maths** gap ≈ **1.8** marks per student · **1** question with a gap of 2+ | | | | |
| **Differential Equations** | 5 | **11.2** vs **9.0** | **≈ 2.2** | **Q12** |
| **Indefinite Integration** | 4 | **13.4** vs **12.7** | **≈ 0.7** | – |
| **Area Under Curves** | 3 | **6.7** vs **6.9** | **−≈ 0.2** | – |
| **Definite Integration** | 5 | **12.4** vs **13.3** | **−≈ 0.9** | – |
| **2. Physics** gap ≈ **0.9** marks per student · **2** questions with a gap of 2+ | | | | |
| **Electrostatics** | 6 | **15.5** vs **13.8** | **≈ 1.7** | **Q33** |
| **Current Electricity** | 2 | **6.4** vs **6.6** | **−≈ 0.2** | – |
| **Kinematics** | 8 | **16.4** vs **17.0** | **−≈ 0.6** | **Q27** |
| **3. Chemistry** gap ≈ **-0.4** marks per student · **2** questions with a gap of 2+ | | | | |
| **Aldehydes & Ketones** | 6 | **18.2** vs **18.2** | **≈ 0.0** | – |
| **Carboxylic Acids** | 4 | **10.0** vs **10.0** | **≈ 0.0** | **Q50** |
| **Alcohols, Phenols & Ethers** | 7 | **21.8** vs **22.2** | **−≈ 0.4** | **Q46** |

## 4 Biggest gaps — question list

Questions where at least 2 more of the institute's top 10 got the right answer than ours, one table per subject. The cropped question for each is in section 5. With 7 of the 10 students shared, at most 3 answers can differ on any question, so the line is set at 2 instead of 4.

### Maths

**1** question with a gap of 2+

| Q | CHAPTER — WHAT IT TESTS | TYPE | INSTITUTE RIGHT | OURS RIGHT · WRONG · SKIPPED | GAP |
|---|---|---|---|---|---|
| **Q12** | **Differential Equations** — Solve a first-order DE and match its constants | Integer · +4 / 0 | **8**/10 | **6**/10 · 4 · 0 | **+2** |

### Physics

**2** questions with a gap of 2+

| Q | CHAPTER — WHAT IT TESTS | TYPE | INSTITUTE RIGHT | OURS RIGHT · WRONG · SKIPPED | GAP |
|---|---|---|---|---|---|
| **Q27** | **Kinematics** — Projection angle that maximises area under the trajectory | Integer · +4 / 0 | **8**/10 | **6**/10 · 4 · 0 | **+2** |
| **Q33** | **Electrostatics** — Flux through a ring — solid angle | Paragraph · +3 / 0 | **7**/10 | **5**/10 · 5 · 0 | **+2** |

### Chemistry

**2** questions with a gap of 2+

| Q | CHAPTER — WHAT IT TESTS | TYPE | INSTITUTE RIGHT | OURS RIGHT · WRONG · SKIPPED | GAP |
|---|---|---|---|---|---|
| **Q46** | **Alcohols, Phenols & Ethers** — Identifying polar aprotic solvents | Integer · +4 / 0 | **5**/10 | **2**/10 · 5 · 3 | **+3** |
| **Q50** | **Carboxylic Acids** — Molar mass of an acid from NaOH titration | Paragraph · +3 / 0 | **9**/10 | **7**/10 · 3 · 0 | **+2** |

### Where our top 10 did better

**Q2** 5 vs 3 right · **Q10** 10 vs 8 right · **Q21** 6 vs 4 right · **Q51** 5 vs 3 right

**How to read this report.** "Institute top 10" = the *Top 10 Students* rows of the institute's question-wise R/W/U sheet for this test (program ranks 1–10 in Pragya G10 (EXI) Achievers, all campuses). "Our top 10" = this campus's top 10 by program rank. A gap of +5 on a question means 5 more of the institute's top 10 got it right than ours.

## 5 Maths – biggest gaps

**1** question, largest gap first
cropped from the question paper

### Q12

**Differential Equations – Solve a first-order DE and match its constants**

Institute top 10: **8 right**, 2 wrong, 0 skipped · Our top 10: **6 right**, 4 wrong, 0 skipped

Maths · Integer · marking +4 / 0

**Gap +2**

12. The differential equation $\left(xy^3\left(1+\cos x\right)-y\right)dx+xdy=0$ represents the curve $\frac{x^2}{ay^2}=\frac{x^3}{b}+x^2\sin x+cx\cos x-d\sin x+k$. Then a+b+c+d is equal to (where 'k' is arbitrary constant).

**Ours wrong:** Dharshan Raj S, Thonta Harshavardhan, Gajavali Anjaneyulu, Yapala Dhanush Reddy

**Ours skipped:** –

## Physics – biggest gaps

2 questions, largest gap first
cropped from the question paper

### Q27

**Kinematics – Projection angle that maximises area under the trajectory**

Institute top 10: **8 right**, 2 wrong, 0 skipped · Our top 10: **6 right**, 4 wrong, 0 skipped

Physics · Integer · marking +4 / 0

**Gap +2**

27. A projectile is thrown with a speed 10 m/s, at an angle θ with the horizontal. For a certain value of angle of projection $\theta = \frac{(n+1)\pi}{12}$ the area bound by the trajectory with horizontal is maximum value of $n$ is (g=10 m/s$^2$). Assume no airdrag.

**Ours wrong:** Dharshan Raj S, Mano Nischal Reddy D, Vihaan Bussari, Gajavali Anjaneyulu

**Ours skipped:** –

### Q33

**Electrostatics – Flux through a ring – solid angle**

Institute top 10: **7 right**, 3 wrong, 0 skipped · Our top 10: **5 right**, 5 wrong, 0 skipped

Physics · Paragraph · marking +3 / 0

**Gap +2**

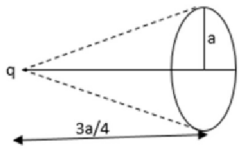
**PARAGRAPH-II**

The electric flux linked with a surface is the number of electric field lines associated with that area. Gauss law helps determine the flux

$$\oint \vec{E}.\vec{ds} = \frac{Q_{inc}}{\varepsilon_0}$$

For an open surface the flux will change depending on the solid angle. Considering this answer the following questions.

33.




The flux associated with the ring due to the charge q is given by Xq/ε$_0$. The value of X is approx.

**Ours wrong:** Poli Karthikeya Reddy, Shiva Sai Jyothiraditya Kokkonda, Vihaan Bussari, Gajavali Anjaneyulu, Yapala Dhanush Reddy

**Ours skipped:** –

## Chemistry – biggest gaps

2 questions, largest gap first
cropped from the question paper

### Q46

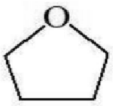
**Alcohols, Phenols & Ethers** – **Identifying polar aprotic solvents**

Institute top 10: **5 right**, 4 wrong, 1 skipped · Our top 10: **2 right**, 5 wrong, 3 skipped

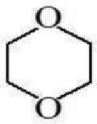
Chemistry · Integer · marking +4 / 0

**Gap +3**

46. How many of the following are polar aprotic solvents?



Diethyl ether, Benzene, Ethyl alcohol,



Petrol

[P] [Q] [R] [S] [T] [U]

Acetone | DMSO (dimethyl sulphoxide | DMF (dimethyl formamide | formic acid

[Q] [W] [X] [Y]

**Ours wrong:** Dharshan Raj S, Thonta Harshavardhan, K Ranveer Reddy, Mano Nischal Reddy D, Vihaan Bussari

**Ours skipped:** Shiva Sai Jyothiraditya Kokkonda, Gajavali Anjaneyulu, Yapala Dhanush Reddy

### Q50

**Carboxylic Acids** – **Molar mass of an acid from NaOH titration**

Institute top 10: **9 right**, 1 wrong, 0 skipped · Our top 10: **7 right**, 3 wrong, 0 skipped

Chemistry · Paragraph · marking +3 / 0

**Gap +2**

**PARAGRAPH-II**

Carboxylic acids are organic compounds containing the carbonyl(>C=O) and hydroxyl (-OH) functional groups combined as a carboxyl group(-COOH). Due to resonance stabilization of the carboxylate anion, carboxylic acids exhibit higher acidity compared to alcohols and phenols. They participate in several characteristic reactions, including nucleophilic acyl substitution (esterification, amide formation, acid halide synthesis), decarboxylation, and reduction to primary alcohols. Furthermore, the presence of the –COOH group strongly influences physical properties, enabling extensive intermolecular hydrogen bonding that results in higher boiling points and stable dimeric structures in non-polar solvents.

50. A 0.600 g sample of a pure monoprotic carboxylic acid (R-COOH) requires 25.00 mL of 0.200 M NaOH for complete neutralization. Calculate the molar mass of the carboxylic acid?

**Ours wrong:** Dharshan Raj S, Vihaan Bussari, Gajavali Anjaneyulu

**Ours skipped:** –